Cambridge (CIE) A Level Computer Science

Your notes

Operating Systems

Contents

- Operating System Fundamentals
- Utility Software
- Program Libraries

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Purpose of an OS

What is an operating system (OS)?

- An operating system (**OS**) is software that **provides an interface between the user and the hardware** in a computer system
- An operating system **hides the complexities of the hardware** from the user, for example:
  - A user **does not need to know** 'where' on secondary storage data is kept, just that it is saved for when they want it again
- To help achieve this an OS provides a **user interface**

User interface

- A user interface is **how the user interacts with the operating system**
- Examples of user interfaces include:
  - **Command Line Interface (CLI)**
  - **Graphical User Interface (GUI)**
  - **Menu**
  - **Natural language (NLI)**

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What is a command line interface (CLI)?

- A Command Line Interface (**CLI**) requires users to interact with the operating system using **text based commands**
- CLIs are more commonly used by **advanced users**
- Examples of CLIs are MSDOS (Microsoft Disk Operating System) and Raspbian (for Raspberry Pi)

What is a graphical user interface (GUI)?

- A Graphical User Interface (**GUI**) requires users to interact with the operating system using **visual elements** such as windows, icons, menus & pointers (**WIMP**)
- GUIs are **optimised for mouse and touch gesture input**
- Examples of GUIs are Windows, Android and MAC OS

What is a menu interface?

- A menu interface is **successive menus** presented to a user with a **single option at each stage**
- Often performed with **buttons** or a **keypad**
- Examples include
  - **Chip and pin machines**
  - **Vending machines**
  - **Entertainment streaming services**

What is a natural language interface (NLI)?

- A natural language interface (**NLI**) uses the spoken word to respond to spoken or textual inputs from a user
- Examples include
  - **Virtual assistants** - Amazon Alexa, Google Assistant, Siri
  - **Search engines**
  - **Smart home devices**

OS management tasks

An operating system has five key **management tasks**:

- **Memory management**
- **File management**
- **Security management**
- **Hardware management**
- **Process management**

Memory management

- Memory management is a process carried out by the operating system **allocating main memory (RAM) between different programs** that are **open at the same time**
- The OS is responsible for **copying programs and data from secondary to primary storage** as it is needed

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- **Programs and data require different amounts of RAM** to operate efficiently and the OS manages this process
- **RAM is allocated** based on **priority and fairness**, for example, system applications (essential) may have a higher priority than user applications
- The OS **dynamically manages the memory**, adjusting allocation as needed to maintain optimal system performance
- Memory management makes **multitasking** possible

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File management

- File management is a process carried out by the operating system **creating, organising, manipulating and accessing files and folders on a computer system**
- The OS **manages where data is stored** in both **main memory** and **secondary storage**
- File management gives the user the ability to:
  - **Create files/folders**
  - **Name files/folders**
  - **Rename files/folders**
  - **Copy files/folders**
  - **Move files/folders**
  - **Delete files/folders**
- The OS allows users to control who can access, modify and delete files/folders (**permissions**)
- The OS provides a **search facility** to find specific files based on various criteria

Security management

- Operating systems provide various security features such as **password-protected system accounts**, a **firewall**, virus scanning and **file encryption**
- Password-protected system accounts are a very common feature in operating systems
- A system administrator is able to **allocate** different **access rights** for different users on a network
- The OS is able to **maintain settings** for individual users, such as desktop backgrounds, icons and colour schemes
- The OS **audits** (keeps a log of) files created by users, accesses, edits and deletes

Hardware management

- Peripheral management is a process carried out by the operating system **managing the way peripherals (hardware) interact with software**
- The OS **allocates system resources** to peripherals to ensure **efficient operation**

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- Peripheral management makes **plug-and-play** (**PnP**) functionality possible, **automatically detecting and configuring new peripherals** without the need for manually installing device drivers or power cycling the system
- A device driver is **a piece of software used to control a piece of hardware**
- Inputs/outputs **require device drivers** in order to be used by the operating system
- The OS has **generic device drivers built in** which makes **basic compatibilit**y possible
- In order for hardware to be used to its **maximum capacity**, often **a separate device driver must be downloaded** from the manufacturer
- Device drivers are **OS specific** and are **regularly updated**

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Process management

- Process management is a process carried out by the operating system **dividing time** (**time slicing**) in to small chunks and **allocating them to different processes**
- The CPU can only execute **one process at a time**, it can can execute **billions of them in one second**
- The OS uses a **scheduling algorithm** to **prioritise** processes
- Processes are **placed in queue** whist waiting to be carried out, they return to the back of the queue when their time is up
- The goal of process management is to **share resources** (**CPU & main memory**)

Worked Example

Identify four key management tasks that the Operating System will perform. **[4]**

Answer

- Memory management [1 mark]
- File management [1 mark]
- Security management [1 mark]
- Hardware / device / peripheral / resources management [1 mark]
- Input/output management [1 mark]
- Process management [1 mark]
- Error checking and recovery [1 mark]
- Provision of a platform for software [1 mark]
- Provision of a user interface [1 mark]

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Utility software

What is utility software?

- Utility software is a **collection of tools** designed to help **maintain** a computer system
- Utility software is designed to **perform a limited number of tasks**
- Utility software **interacts with the computers hardware**, for example, secondary storage devices
- Some utility software comes **installed with the operating system**
- Examples of utility software include:
  - **Disk formatter**
  - **Virus checker**
  - **Defragmentation software**
  - **Disk contents analysis/disk repair software**
  - **File compression**
  - **Back-up software**

Disk formatter

- A disk formatter **prepares a storage device** (like a hard drive or USB stick) for use by **creating a file system** and **organising the space into sectors and tracks**
- It is needed:
  - To **wipe** and **re-initialise** a disk before use
  - To change the **file system format** (e.g. FAT32, NTFS)
  - To **remove all data** and errors before installing a new OS or reusing the drive
- For example:
  - **Formatting** a new external drive before saving files
  - **Reformatting** a corrupted USB stick to make it usable again

Virus checker

- Virus checking software is a **combination of different software to prevent** computers from being susceptible to **viruses** and other **malicious software**
- It typically includes:
  - **Anti-virus**

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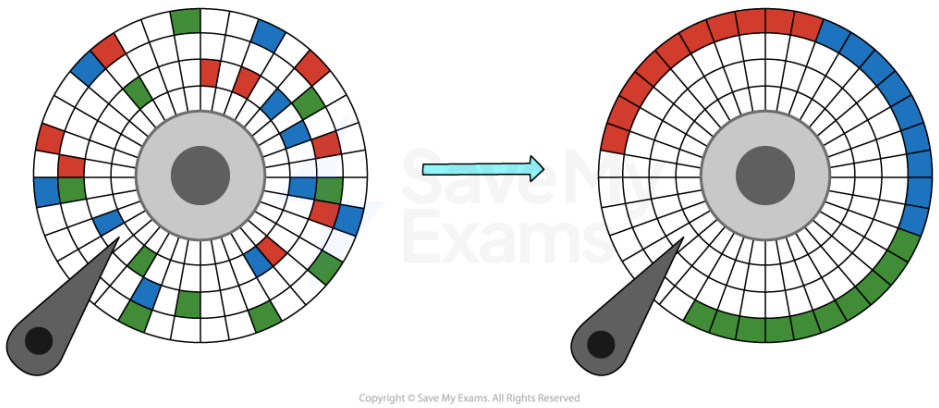
- **Anti-spam**
- **Anti-spyware**

- A virus checker **scans** through **email** attachments, **websites** and downloaded **files** to search for issues
- Two ways virus checkers can approach the task are:
  - Use a list of known **unique malware fingerprints** (**signatures**) to block immediately if they try to access your device in any way
  - **Monitor** the **behaviour** of programs to identify suspicious activities that might indicate malware such as;
    - **Rapid deletion/modification of files**
    - **Attempts to access sensitive data/resources**
    - **Communicating with known malicious servers**
- Virus checkers will also perform **checks for updates** to ensure the database of signatures is **up to date**

Defragmentation software

- Defragmentation software **groups fragmented files back together** in order to **improve the access speed** of a hard disk drive **(HDD)**
- As programs and data are added to a new hard disk drive, it is added in order, **over time as files are deleted this leaves gaps**
- As programs and data are added over time, **these gaps get filled and data becomes fragmented**
- Defragmentation can only be used on **magnetic storage**



Disk contents analysis/disk repair software

- This utility checks the **structure and health** of a disk, including **files, folders, sectors**, and the **file allocation table**

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- It is needed:
  - To **diagnose and fix errors** on the disk (e.g. bad sectors, file system corruption)
  - To **recover lost or damaged files**
  - To improve system performance by identifying and repairing issues
- For example:
  - Running **Check Disk (chkdsk)** on Windows to repair file system errors
  - Using a **disk utility** on macOS to verify and repair drive problems

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File compression

- Compression **reduces the amount of secondary storage** required by performing an algorithm on the original data
- **Lossy compression** physically **removes data** from the original data to reduce its size, **the original file can not be re-created**
- **Lossless compression** uses mathematics to **order data** more efficiently reducing its size, **the original files can be re-created as no data is lost**

Back-up software

- Back-up software is used to **create copies of personal data** in order to keep it safe in the event of:
  - **Accidental loss**
  - **Data theft**
- Backups can be **automated** and **scheduled** to happen at less busy periods of the day, to not take up valuable system resources (e.g. overnight etc.)
- Backups can be made in two ways:
  - **Full** - all files are backed up (saftest, slow)
  - **Incremental** - only files that have been added/modified since the last backup are backed up (faster, less secure)
- Backups can be stored **locally** (secondary storage) or **remotely** (cloud)
- Backup software can be **purchased** or come as a standalone application **bundled** with an **operating system**

Worked Example

Draw one line from each utility software to its correct description.

Utility software Description

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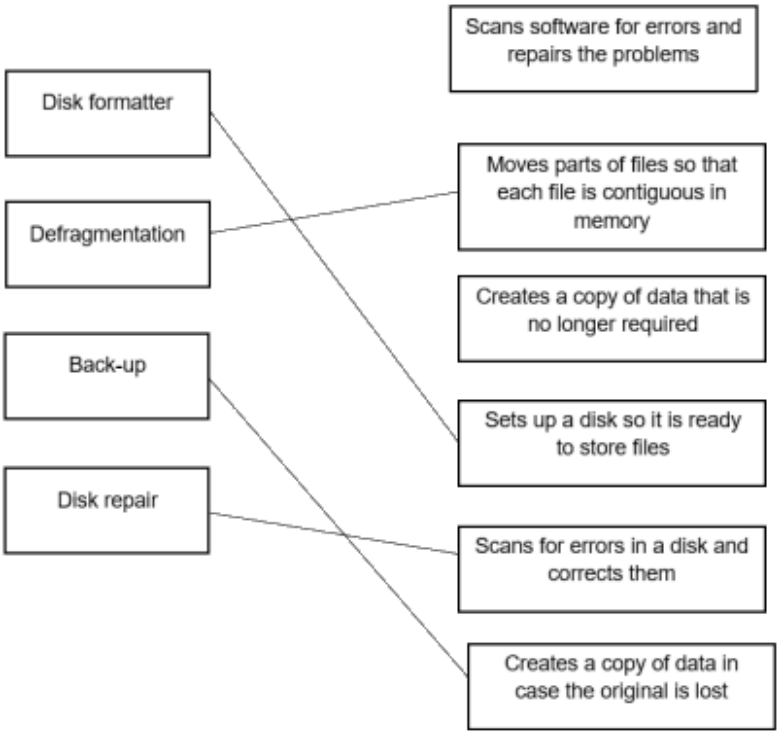
	Scans software for errors and repairs the problems
Disk formatter	Moves parts of files so that each file is contiguous in memory
Defragmentation	Creates a copy of data that is no longer required
Back-up	Sets up a disk so it is ready to store files
Disk repair	Scans for errors in a disk and corrects them
	Creates a copy of data in case the original is lost

[4]

Answer



Your notes

Program libraries

What is a program library?

- A program library is a **collection of pre-written code** (called **library routines**) that programmers can **reuse** in their own software
- These routines can **perform common tasks** such as:
  - **Sorting**
  - **Displaying graphics**
  - **Playing sounds**
  - **Managing data**
- Advantages of using program libraries includes:

Advantage	Explanation
Saves development time	Pre-written routines can be reused, so programmers don't need to write common functions from scratch
Promotes code reuse	Well-tested routines can be used in multiple programs, reducing duplication
Improves reliability	Library routines are usually thoroughly tested and debugged, reducing the chance of errors
Easier maintenance	Updates to a library routine automatically benefit all programs that use it (especially with **DLLs**)
Efficient memory usage	**Dynamic link libraries (DLLs)** are only loaded when needed at runtime, saving system resources
Standardisation	Promotes consistency in how common tasks (e.g. file handling, sorting) are implemented

What is a DLL?

- A **Dynamic Link Library (DLL)** is a file that contains **pre-written code**, such as **functions or routines**, that can be **used by multiple programs** at **runtime**

Aspect	Advantages	Disadvantages

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Program Size	Reduces program size by keeping reusable code separate	-
Memory Usage	Saves memory by sharing DLLs between programs	-
Code Reuse	Encourages reuse of common routines (e.g. sorting, sound)	-
Modularity	Supports modular design for easier development and maintenance	-
Ease of Updates	Update a DLL once to improve all programs that use it	Can cause **version conflicts** if newer DLLs aren't compatible
Dependencies	External DLLs can be reused by many applications	Program may **fail to run** if a required DLL is **missing**
Security	-	Malicious or altered DLLs can pose **security risks**
Debugging	-	Errors in DLLs can be **hard to trace** across multiple programs

Your notes

Worked Example

Jennifer is writing a computer program for her A Level homework.

Jennifer uses a program library to help her write her computer program.

Describe how a program library can be used while writing a computer program. **[2]**

Answer

- Program libraries store pre-written functions and routines [1 mark]
- The program library can be referenced/imported [1 mark]
- The functions/routines can be called in her own program [1 mark]